

Empirical Proof Record: VALIDATED

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Proof ID: 7f6a839eb599a92c43a7aafa736a0747...

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1. Claim

Kimera distinguishes the minimum-drawdown and maximum-drawdown orderings of the same real return multiset (genuinely different risk, identical trade-set), where a set-based retriever represents them identically: $\text{history_separation_advantage} = \text{mean}(\text{kimera_prime_distance}) - \text{mean}(\text{rag_representation_distance}) > 0$, with Kimera separating every pair and the RAG side conflating every pair.

- **Operationalisation:** For 5 real-market return multisets, find the min-DD and max-DD ordering (search of 300 shuffles + sorted extremes). Run both through Kimera entity · batch (fresh trajectories). $\text{kimera_prime_distance} = 1 - \text{Jaccard}(\text{prime_chain})$. $\text{rag_representation_distance} = \text{mean-pooled TF-IDF divergence} (\approx 0, \text{ same multiset})$. VALIDATED iff advantage > 0 with separation rate 1.0 (each $\text{kimera_prime_distance} \geq 0.5$, each rag ≈ 0).
- **Threshold:** $\text{history_separation_advantage} > 0.0$ $\text{representation_distance}$
- **H0:** H0: advantage ≤ 0 — Kimera conflates the min-DD and max-DD orderings no better than a set-based retriever
- **H1:** H1: advantage > 0 — Kimera separates real-risk-different same-set histories that a set-based retriever conflates

2. Verdict

VALIDATED — observed 1.0

$\text{history_separation_advantage} + 1.0000 = \text{Kimera prime-distance } 1.0000 - \text{RAG distance } 0.000000$ over 5 real min-DD vs max-DD pairs; separation rate 1.00 (Kimera distinguishes real-risk-different same-set histories, RAG conflates). Mean

real ΔDD 0.0458 (the histories Kimera separates are genuinely different risk).
 NOTE: this is discrimination, not magnitude — Kimera does NOT grade by
 drawdown (MEM4 $\rho=+0.18$ ns; diagnostic $\rho\approx 0$). Series FRED SP500 (real
 market)

3. Evidence

Pillar	Statistic	Value	95% CI
0.comparison.finance_history_discrimination	history_separation_advantage	1.0	—

4. Pre-registration

- Registered at: 2026-05-22T19:57:14.638121+00:00
- Config hash: a337cd052a420c174c18f3dfef589393...
- Data hash: 4ff30d839032219673d615128ef6cf58...

5. Signature

26d47a913eed1b7c5c5195c11d6da1bfe878225a4dce1138638316ddf1961651